the properly calibrated GAMMADIST results against the output from your favorite financial planning software. Send your thanks to Bill Gates and his team at Excel.

The Effect of Asset Allocation on Sustainability

Clearly, the odds of earning a specific return depend on your asset allocation; that is, on how you divide your capital between stocks, bonds, and other investments.

The technical question I want to investigate using the simple formula I introduced in the previous section is, "How does the asset allocation affect the probability that you will be able to consume $16,000 annually for the rest of your life?"

Let's return to the original example in which a 65-year-old wishes to withdraw 8% per year. Tables 8.4 to 8.6 assume average male and female mortality and provide insight into the effect of asset allocation. Table 8.4 tells us, for example, that if a 65-year-old allocates all of his $200,000 to a short-term bank account or money market fund, earning an annual rate of 1.3% (after inflation), and has zero exposure to the equity markets, there is only a 31.1% chance that this person will be able to consume the desired $16,000. In other words, with a real consumption level of $16,000, there's a 68.9% chance that this person will outlive his money. You will likely agree that this is a very low probability of success. I personally would like to see numbers that are north of 90%, although I could live with numbers as low as 80% if there were some flexibility in my retirement spending budget.

The only difference among Tables 8.4, 8.5, and 8.6 is the age at which the analysis is being conducted. Notice that the older you are when you retire—all else being equal—if you are planning on spending the same amount of money, the sustainability odds are better. This is because, perversely, you have less time ahead of you.

Another byproduct of this sort of analysis is that it sheds light on the affordability, or lack thereof, of early retirement.

Now, what happens if you allocate your assets to funds with somewhat greater volatility but higher long-term growth rates? The